

Query Processing in Spatial Network Databases

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Outline

- Introduction
- Related work
 - Disk-based graph representations: 2DMatrix, CCAM structure
 - R-tree
- Spatial query in network databases
 - Architecture
 - Spatial queries:
 - Nearest neighbor query
 - Incremental Euclidean Restriction (IER) method
 - Incremental Network Expansion (INE) method
 - Range query
 - Range Euclidean Restriction (RER) method
 - Range Network Expansion (RNE) method
- Experiments
- Summary

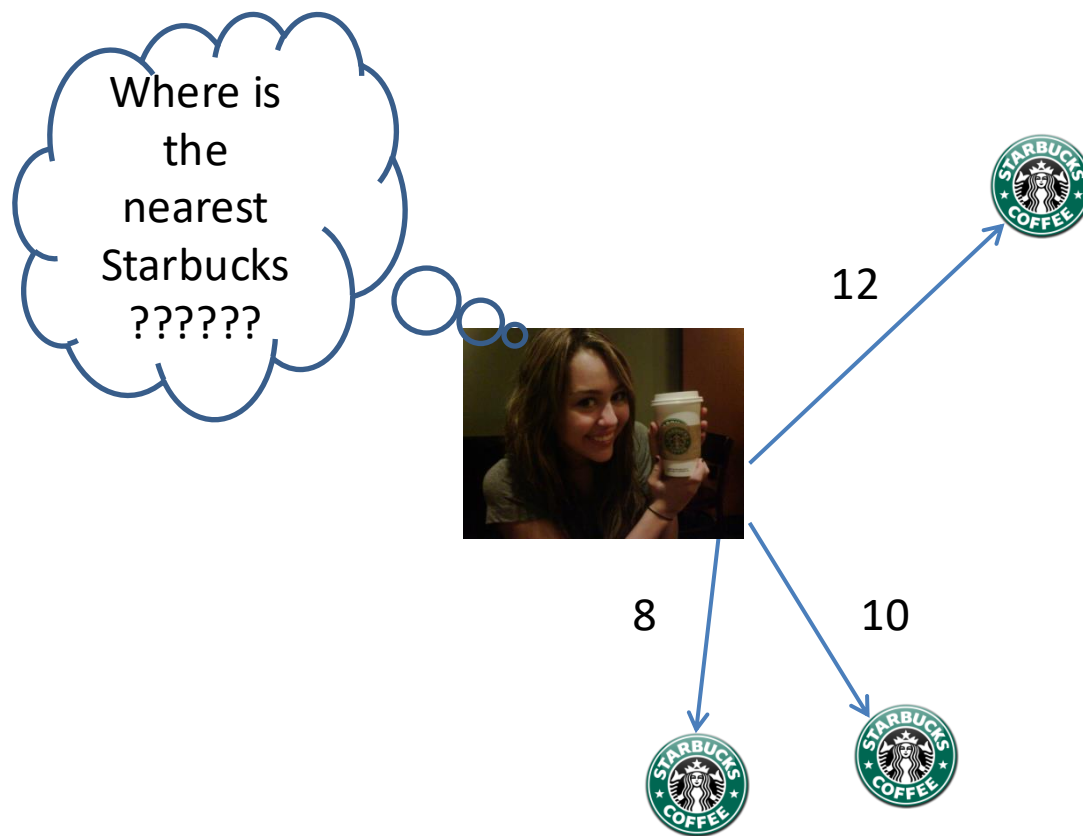
Introduction

■ Euclidean distance



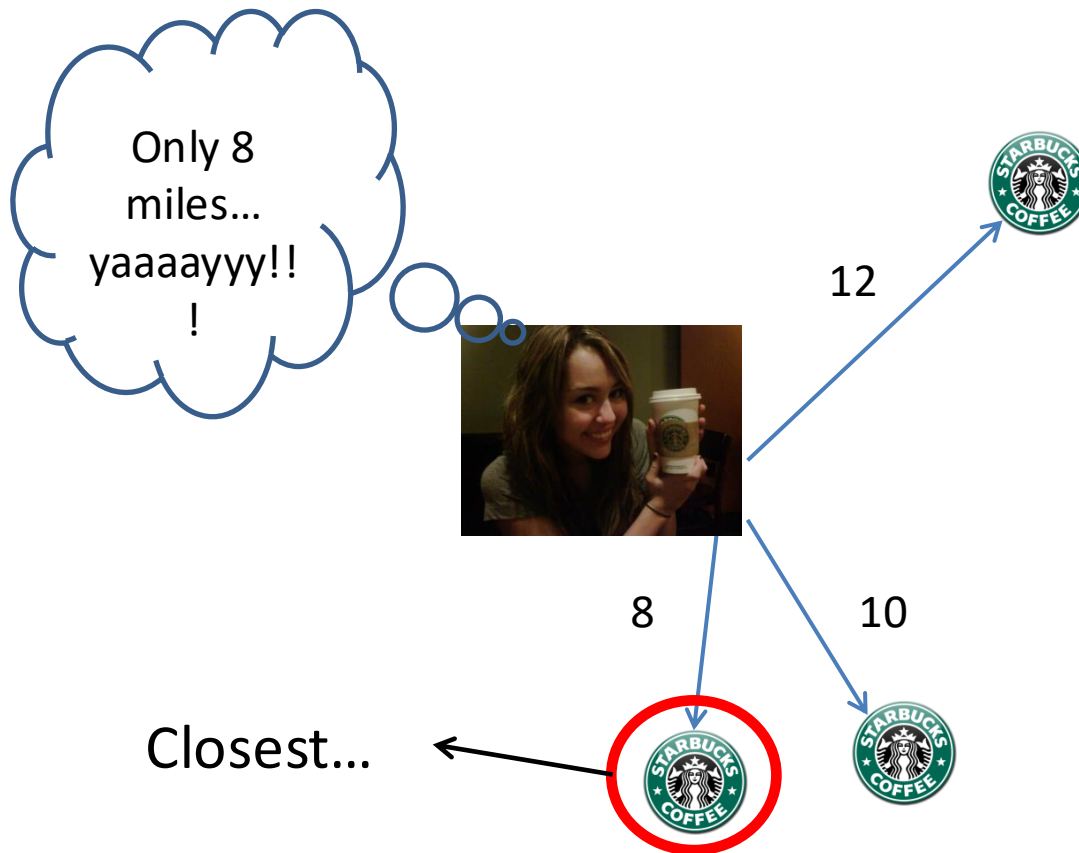
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■ Euclidean distance



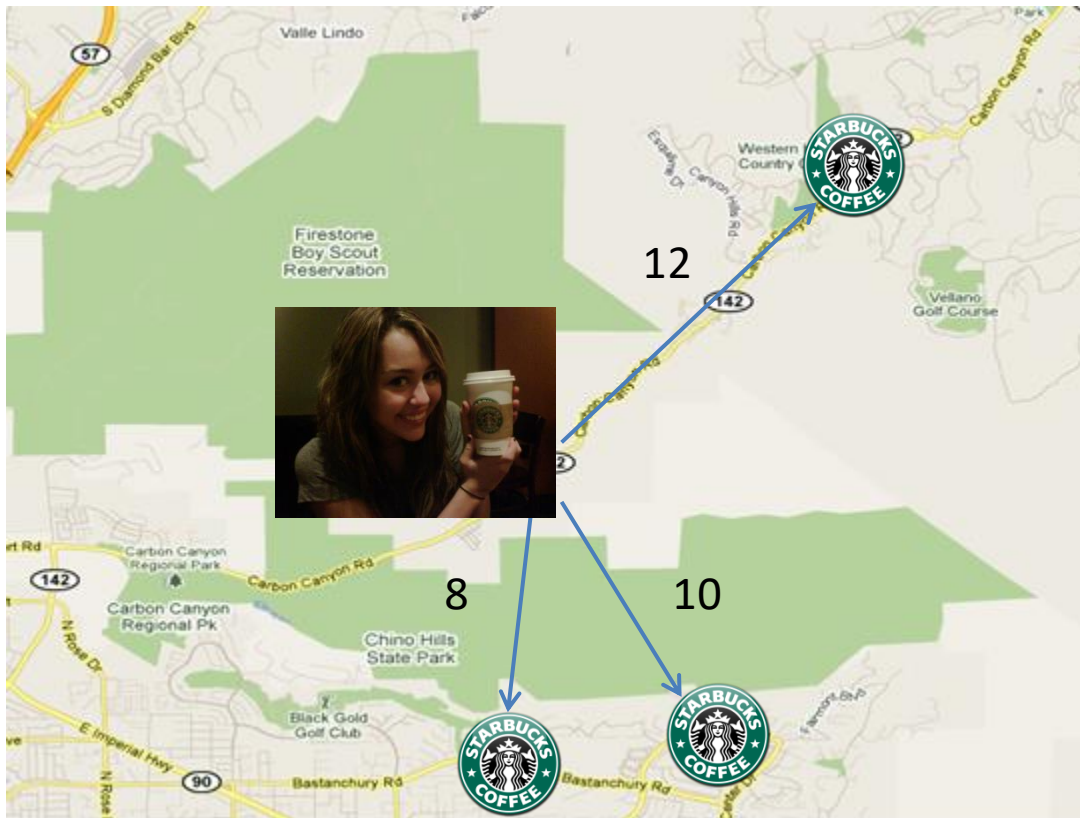
Introduction

■ Euclidean distance



Introduction

- Network distance

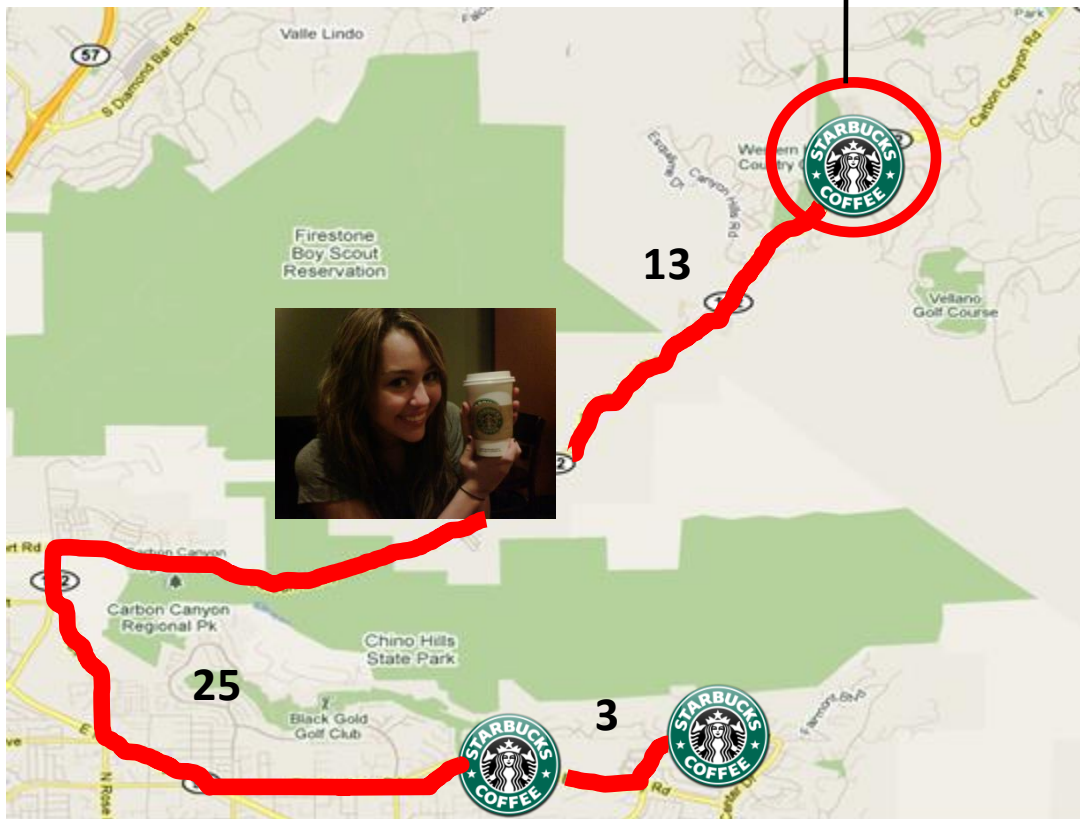


Considering the underlying road network, **is the nearest Starbucks still the same**



Introduction

- Network distance



NO !!!

Introduction

- Euclidean distance vs. Network distance

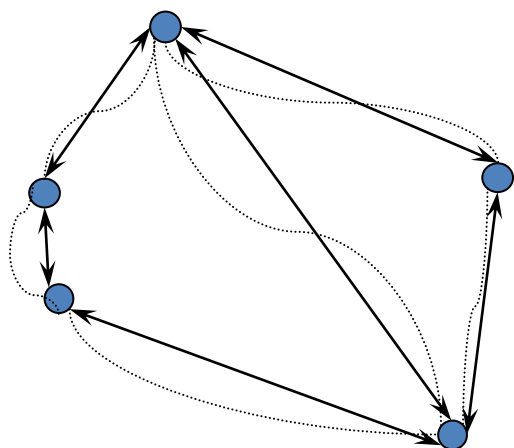
Any relationship?

$$d_E \leq d_N$$

the Euclidean distance between two points is equal or smaller than their network distance.

Introduction

- How can we represent a road network?
- Graphs



Outline

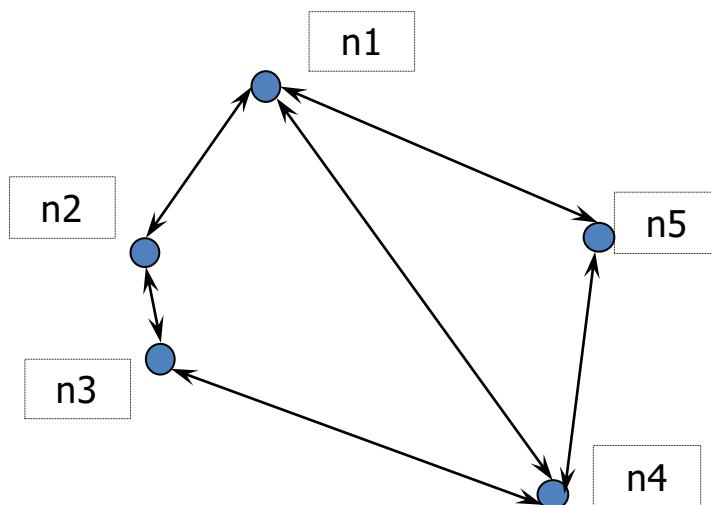
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Disk-based graph representations

- A graph can be represented as
 - Two-dimensional matrix
 - An adjacency list

Disk-based graph representations: 2D Matrix

- Two-dimensional matrix



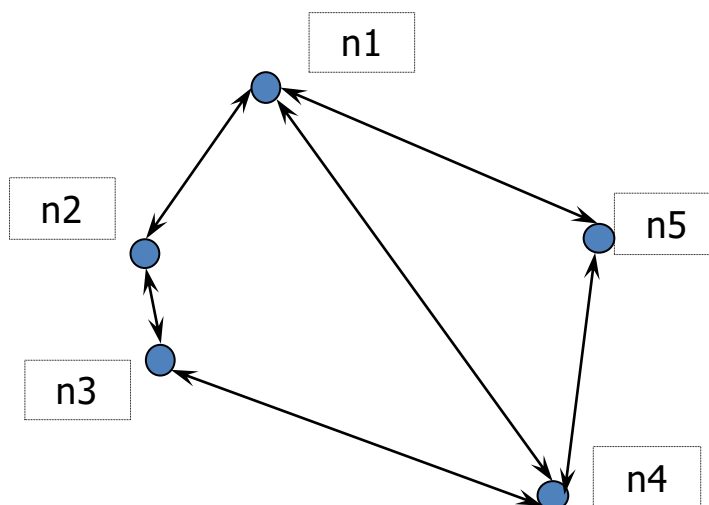
- Disadvantage?

Adjacency matrix

	n1	n2	n3	n4	n5
n1	0	1	0	1	1
n2	1	0	1	0	0
n3	0	1	0	1	0
n4	1	0	1	0	1
n5	1	0	0	1	0

Disk-based graph representations: 2D Matrix

- Two-dimensional matrix



- Disadvantage?
Sparse, More I/O

Adjacency matrix

	n1	n2	n3	n4	n5
n1	0	1	0	1	1
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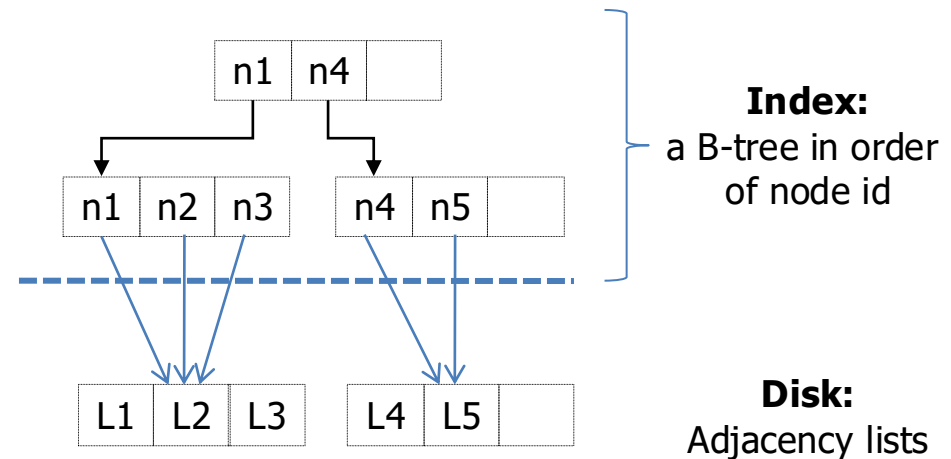
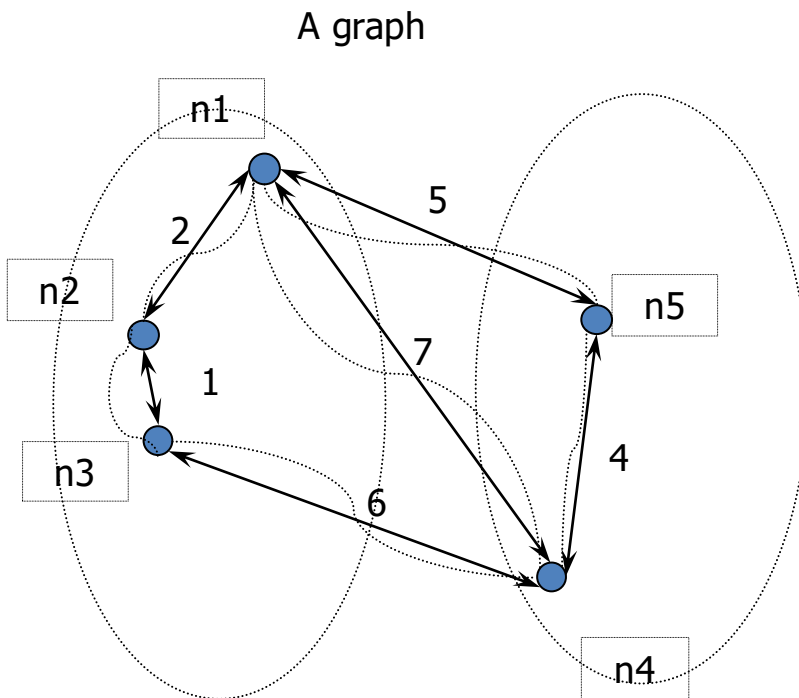


Disk-based graph representations: CCAM structure

- The Connectivity-clustered Access Method (CCAM) structure
 - Each node has a list that stores its neighbors
 - Stores the lists of neighbor nodes together

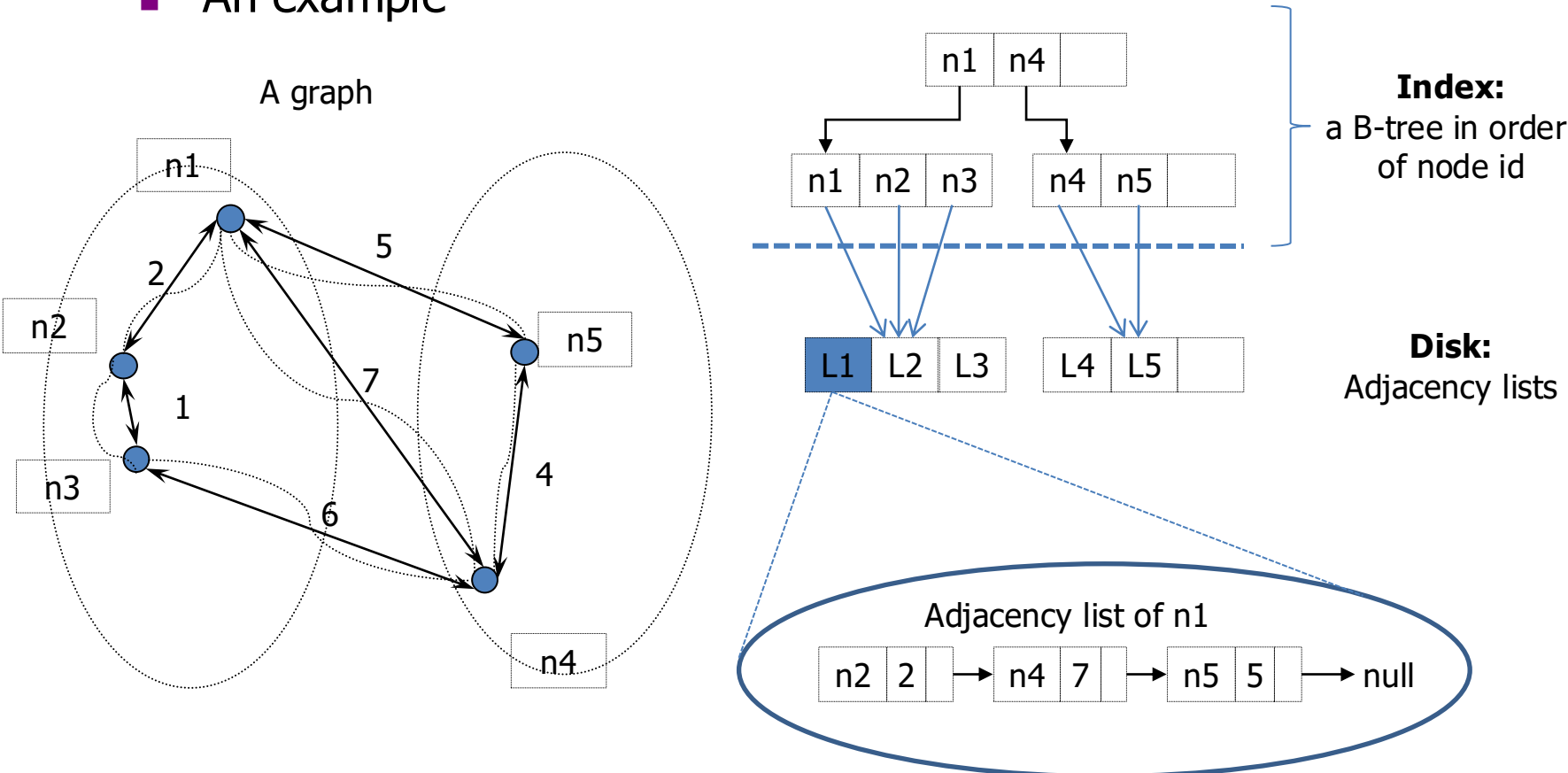
Disk-based graph representations: CCAM structure

- An example



Disk-based graph representations: CCAM structure

■ An example





Disk-based graph representations- CCAM structure

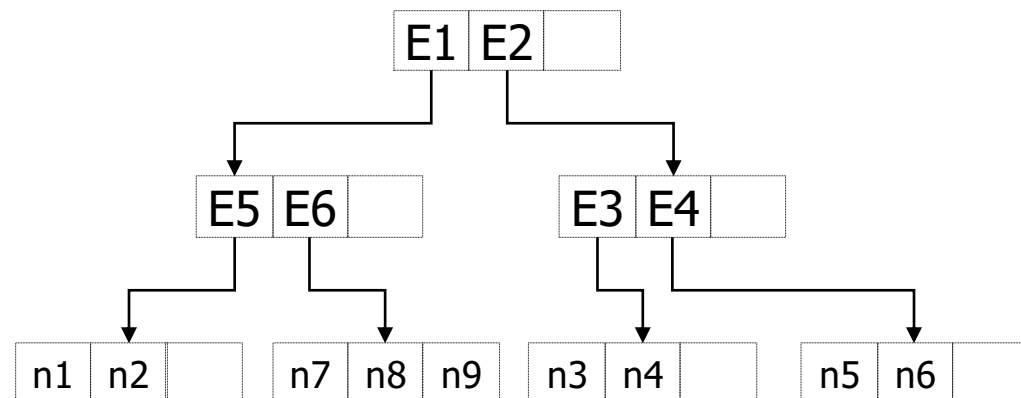
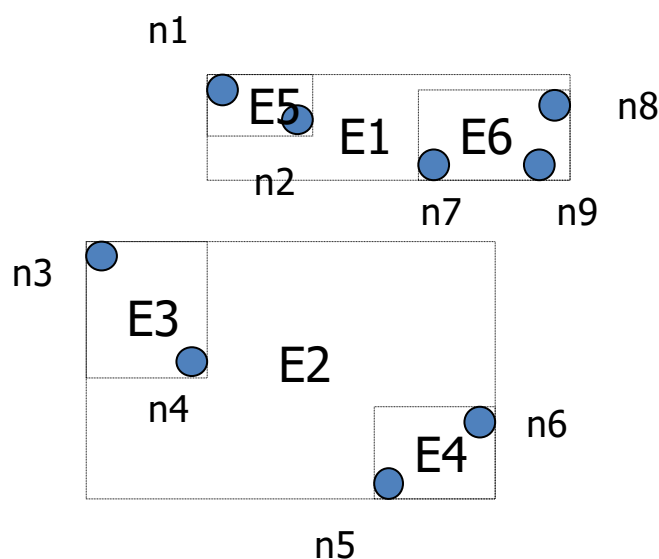
- CCAM vs. 2D Matrix ?
- CCAM is preferable for applications, such as road networks, where the graphs are sparse.

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Spatial query processing in Euclidean Space

- An R-tree index
 - Multidimensional extension of B-tree

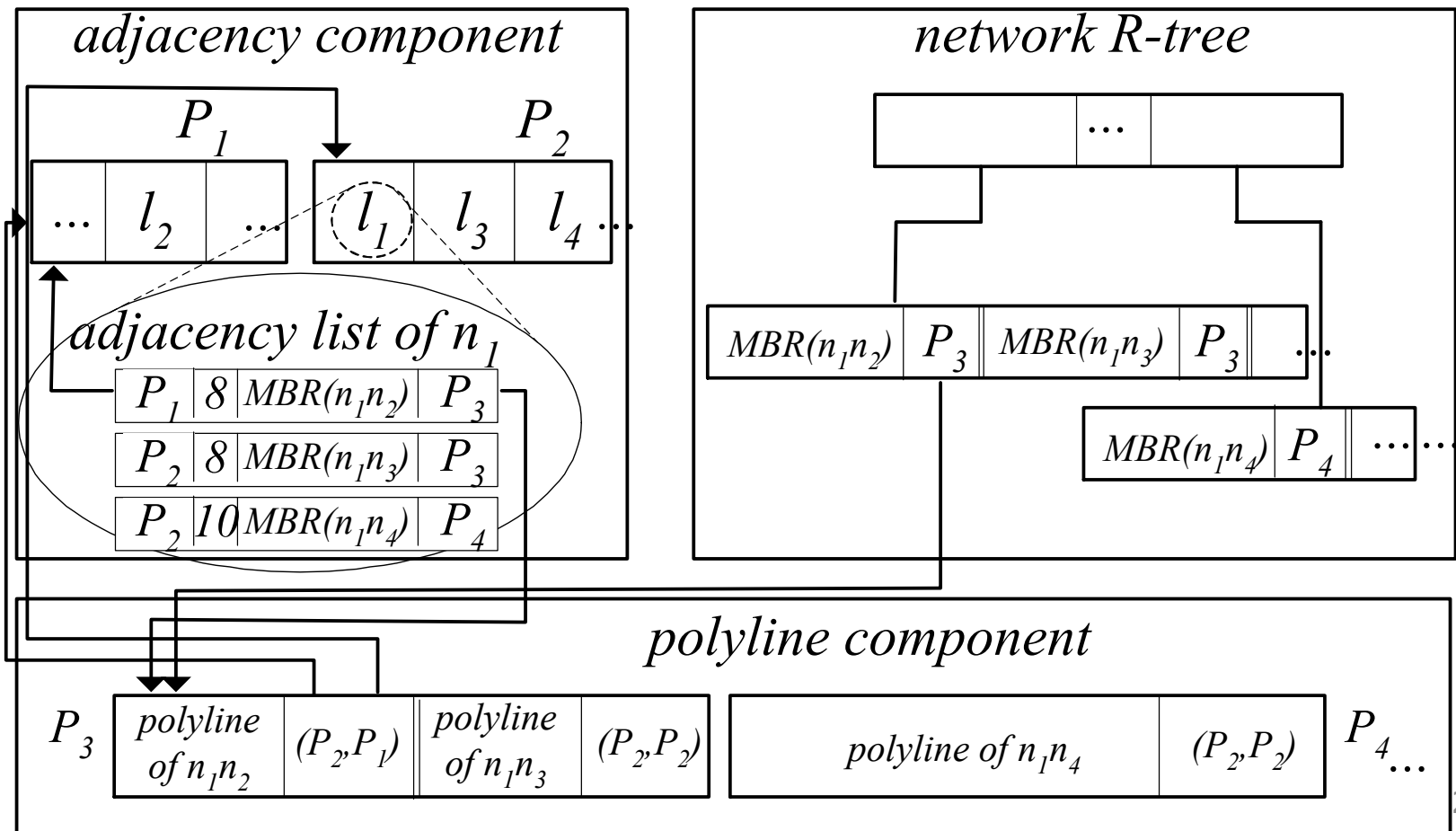


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Architecture

- Underlying network and spatial entities are separated
- Index the entity datasets (e.g., hotels) separately by R-trees (called Objects R-tree)
- For the network: preserve connectivity (adjacency comp) & location (network R-tree)

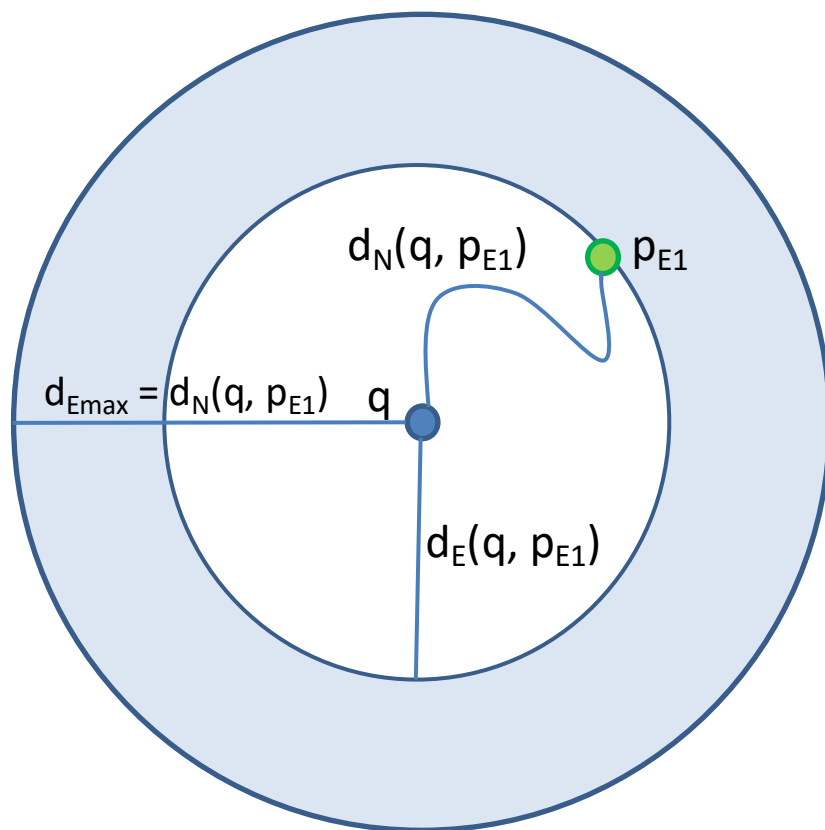


Functions

- *check_entity(seg, p)*: is a Boolean function that returns true if point (entity) p lies on the network segment *seg* (i.e., *seg* covers p)
 - MBR of *seg* is used for filtering and its poly-line representation for refinement.
- *find_segment(p)*: outputs the segment that covers point p by performing a point location query on the network R- tree.
- *find_entities(seg)*: returns entities covered by segment *seg*.
 - first finds all the candidate entities that lie in the MBR of *seg*, and then eliminates the false hits using the poly-line of *seg*.
- *compute_ND(p1, p2)*: returns the network distance of two arbitrary points $p1$, $p2$ in the network,
 - by applying a (secondary-memory) Dijkstra's algorithm to compute the shortest path from $p1$ to $p2$.

Nearest Neighbor - IER

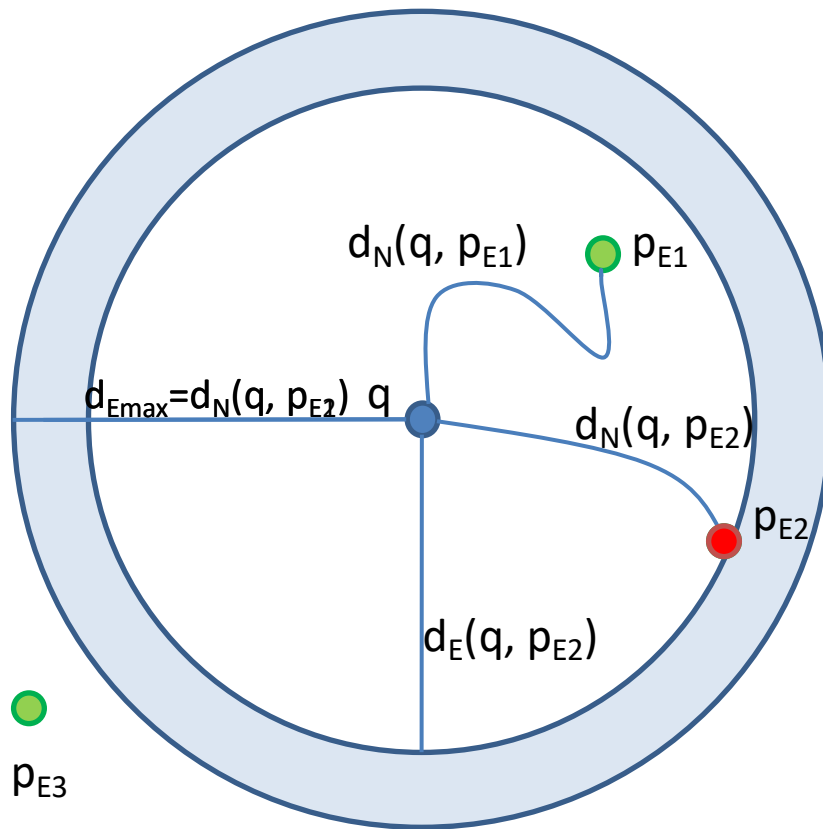
- Example: NN of q ?



- Step 1: Find Euclidean NN p_{E1} of q on the entity R-tree
- Step 2: Compute the network distance $d_N(q, p_{E1})$ of p_{E1}
- Step 3: **Euclidean lower-bound.** Objects closer to q than p_{E1} should be within Euclidean distance $d_{Emax} = d_N(q, p_{E1})$ from q . Only check SHADED AREA!

Nearest Neighbor - IER

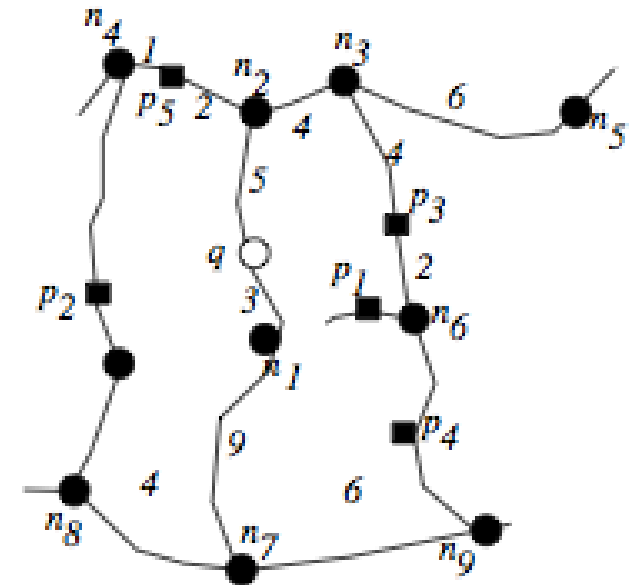
- Example: NN of q ?



- Step 1: Find Euclidean NN p_{E1} of q on the entity R-tree
- Step 2: Compute the network distance $d_N(q, p_{E1})$ of p_{E1}
- Step 3: **Euclidean lower-bound.** Objects closer to q than p_{E1} should be within Euclidean distance $d_{E\max} = d_N(q, p_{E1})$ from q . Only interesting if in SHADED AREA!
- Step 4: Find 2nd Euclidean NN p_{E2} within range $d_{E\max}$.
- Step 5: $d_N(q, p_{E2}) < d_N(q, p_{E1})$. Current NN is p_{E2} .
- Step 6: Set $d_{E\max} = d_N(q, p_{E2})$. Range is now smaller.
- Step 7: Next NN p_{E3} is out of $d_{E\max}$ range. Terminate!

IER: Shortcoming

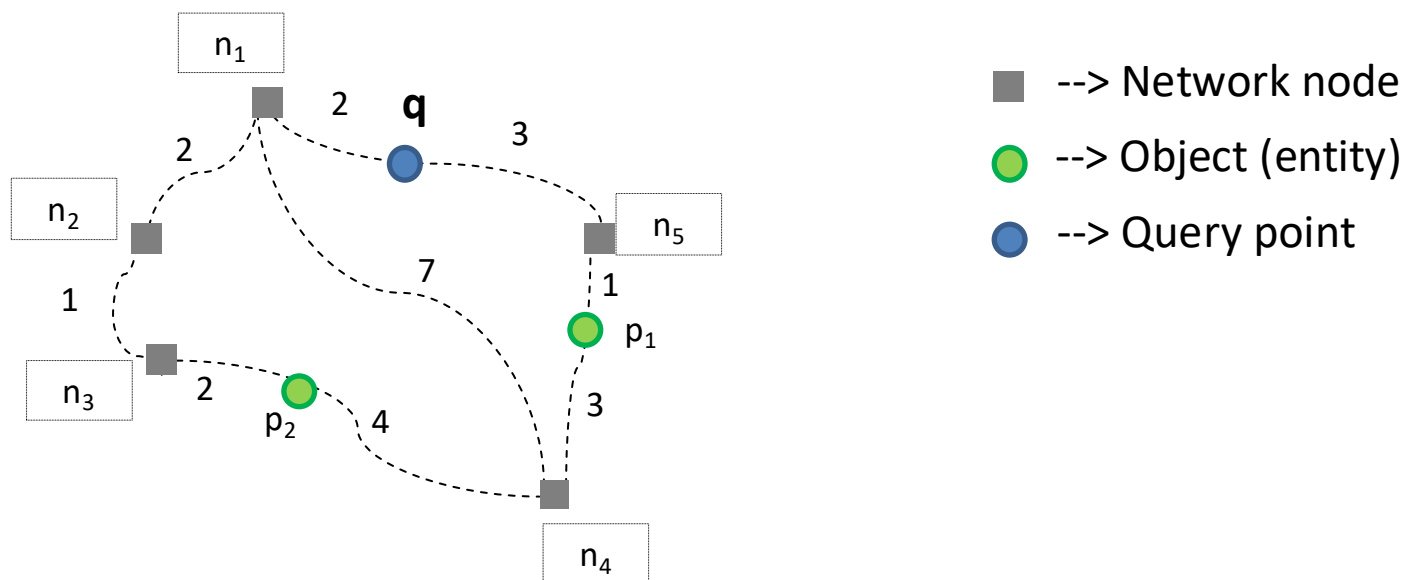
- IER performs well if the ranking of the data points by their Euclidean distance is similar to that with respect to the network distance. Otherwise, a large number of Euclidean NNs may be inspected before the network NN is found
- E.g., P5 is the closest in network distance but farther than p1 to p4 in Euclidean distance



Since p_5 has the largest Euclidean distance, it will be examined after all other entities, i.e., p_1 to p_4 correspond to *false hits*, for which the network distance computations are redundant.

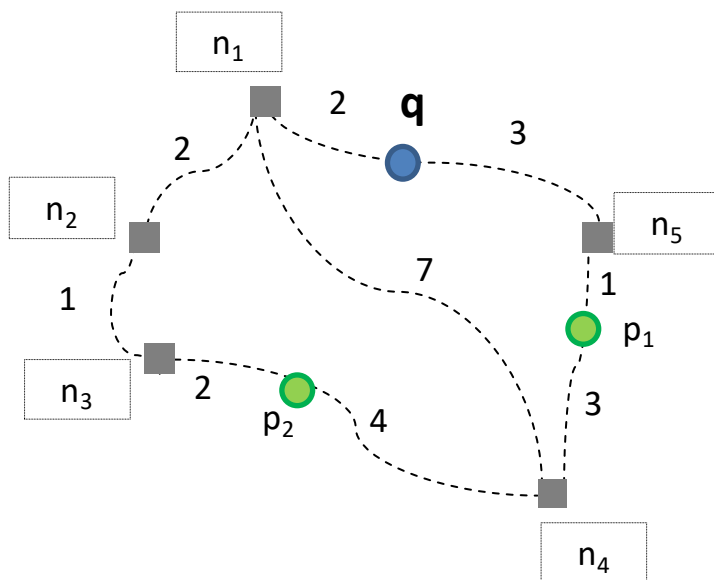
Nearest Neighbor – INE

- Incremental Network Expansion (INE) performs network expansion (starting from q), and examines entities in the order they are encountered.



Nearest Neighbor – INE

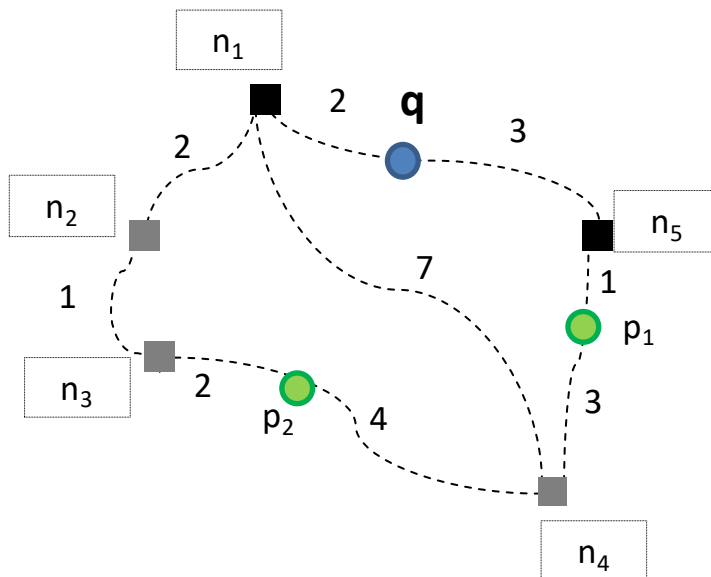
Find NN for a given query point q .



Min Heap

Nearest Neighbor – INE

Find the edge that q is on using Network R-tree, initiate heap

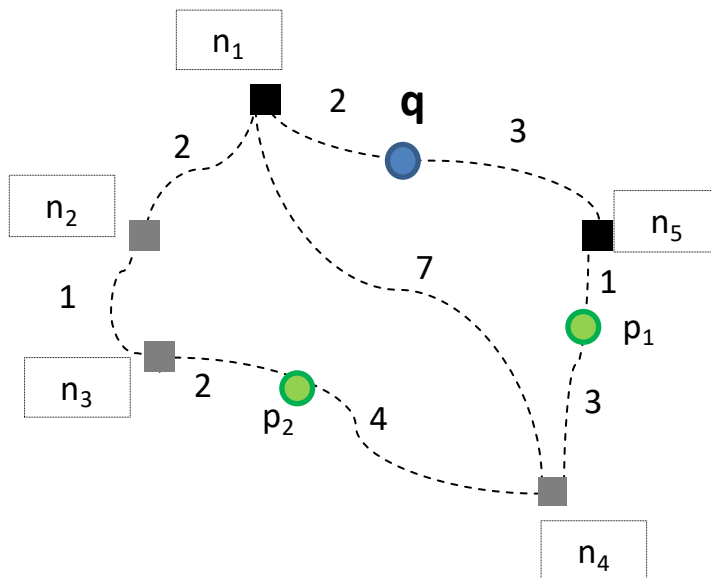


Min Heap

$(n_1, 2), (n_5, 3)$

Nearest Neighbor – INE

For each edge e in heap, find entities that are on e and add e 's neighboring edges into heap.

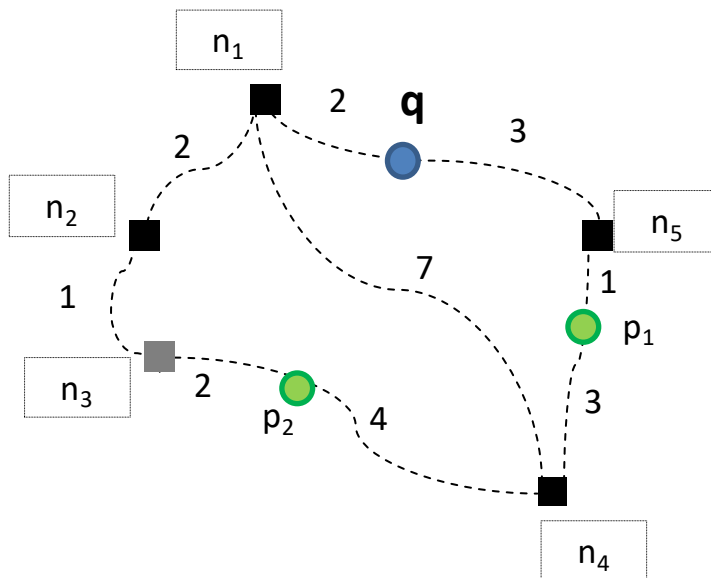


Min Heap

$(n_1, 2), (n_5, 3)$

Nearest Neighbor – INE

Add n_2 and n_4 into heap

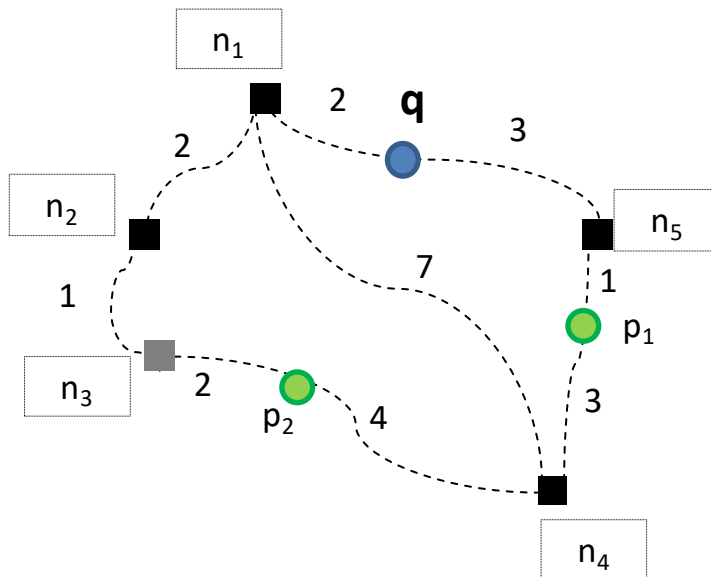


Min Heap

$(n_5, 3), (n_2, 4), (n_4, 9)$

Nearest Neighbor – INE

Add n_4 into heap

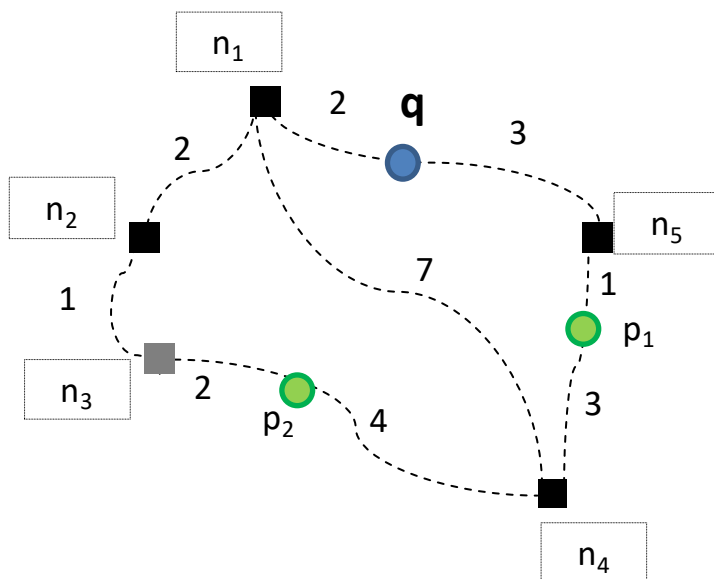


Min Heap

$(n_2, 4), (n_4, 7), \text{---}(n_4, 9)$

Nearest Neighbor – INE

p_1 is found on the edge (n_5, n_4)



- $d_N(q, p_1) = d_{N_{\max}} = 4$
- The next entry in the heap n_2 doesn't have a smaller distance than $d_{N_{\max}}$. Thus, the algorithm **terminates**.

Min Heap

$(n_2, 4), (n_4, 7)$

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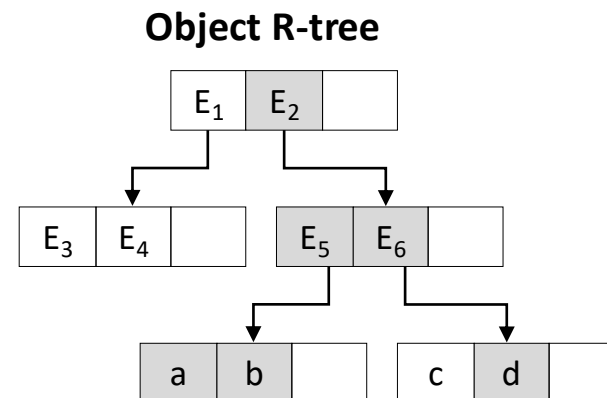
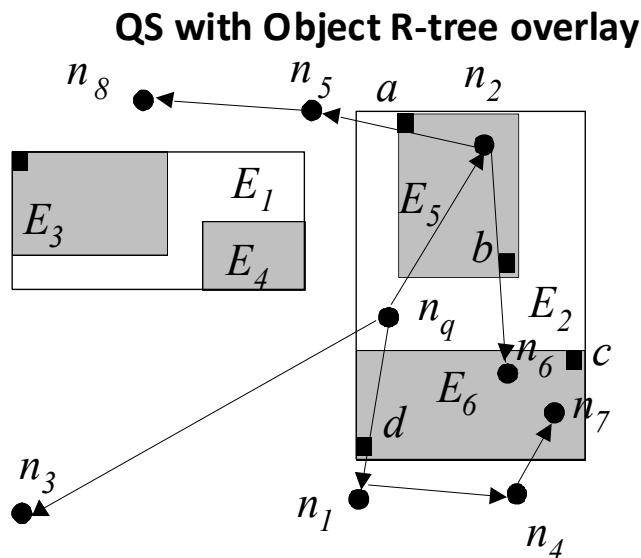
Range Queries – ER

- Range query: given a query point q , a range e and a spatial dataset S , find all objects that are within network distance e from q .
- Range Euclidean Restriction (RER) method:
 - Perform a range query at the entity dataset and find the set of objects S' within (Euclidean) distance e from q .
 - S' is guaranteed to avoid **false misses** using lower-bound property ($d_N(q,p) \leq e \Rightarrow d_E(q,p) \leq e$).
 - But it may contain a large number of **false hits**.
 - RER examines all segments within network distance e from q . Points of S' that fall on some segment, are removed from S' and returned to the user as result.
 - The process terminates when all the segments in the range are exhausted, or when S' becomes empty.

Problem? Works well when S' gets empty first
(i.e., when too many qualified segments, QS)

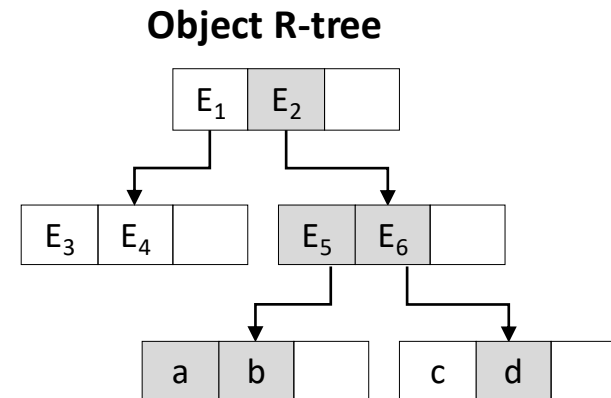
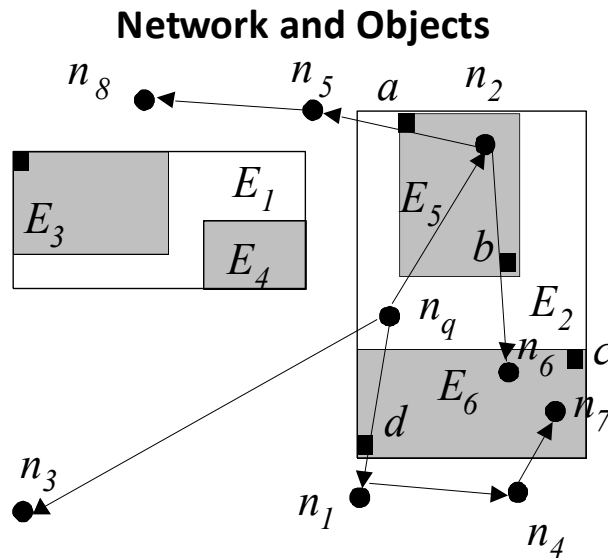
Range Queries – NE

- Range Network Expansion (RNE) algorithm
 - First compute the set QS of qualifying segments within network range e from q .
 - Retrieve the data entities falling on these segments (intersection-join between QS and objects)



Problem? QS may be large

Range Queries – NE - Optimized



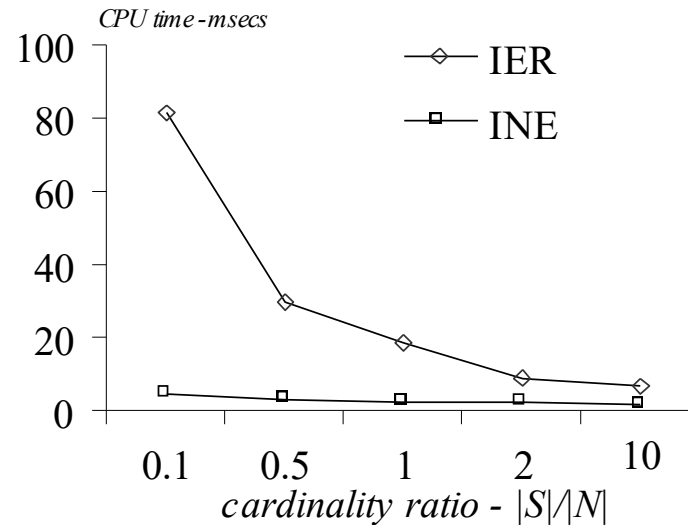
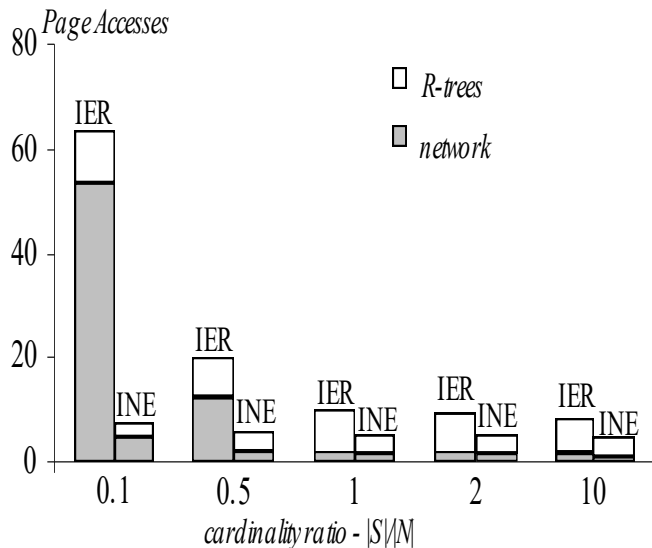
- Start traversing the object R-tree from root
- Compute QS_i for each entry E_i in the current R-tree node.
 - Ex: $QS_1 = \{\}$, $QS_2 = \{\text{all segments except } n_5n_8 \text{ and } n_1n_4\}$, $QS_5 = \{n_qn_2, n_2n_5, n_2n_6\}$
 - For each entry E_i
 - If $QS_i \neq \{\}$, recurse down the tree
- If current node is a leaf (suppose E_6), its points only be checked against QS_6 .
 - Return entities falling on the segments in QS_i (QS_6). Note that c is not a qualifying object.

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Experiments - NN queries

- IER (Incremental Euclidean Restriction) vs. INE (Incremental Network Expansion).
- Cost as a function of the ratio entity/edge cardinality
- Number of neighbours to be retrieved $k=10$

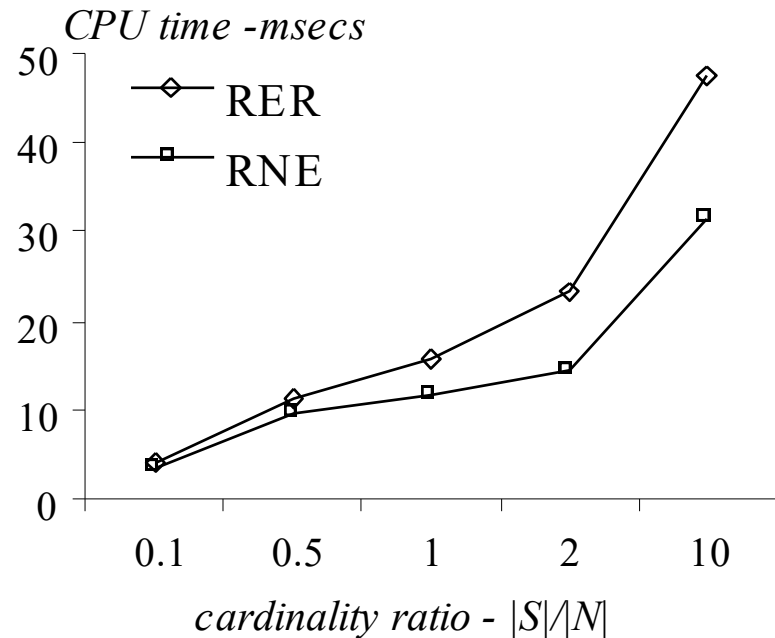
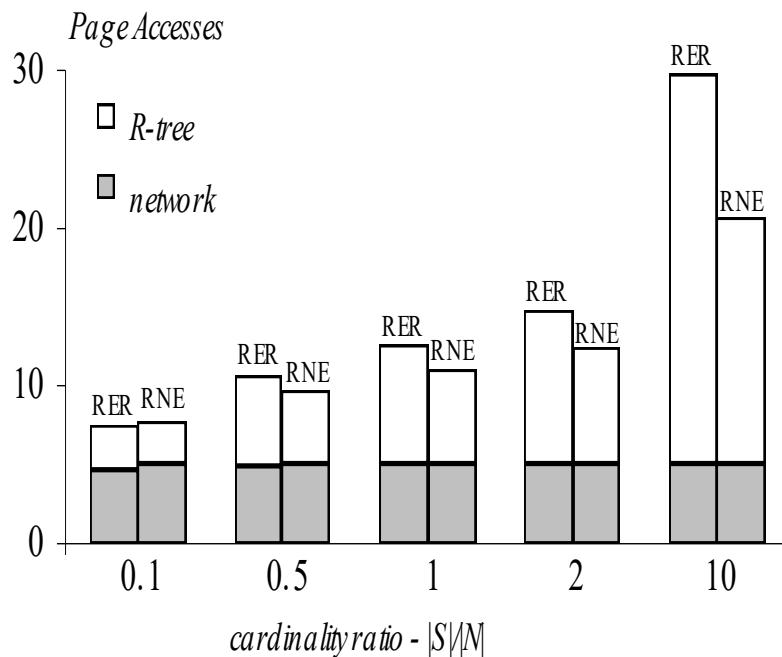


IER: When $|S|$ is small, the Euclidean NNs are far from the query point, which increases the number of false hits and the unnecessary network distance computations.

INE: Low I/O because the range queries on the R-tree exhibit high locality. Moreover, only the necessary network edges are visited (as ensured by the algorithm).

Experiments - Range Search

- RER (Range Euclidean Restriction) vs. RNE (Range Network Expansion).
- Cost as a function of the ratio entity/edge cardinality
- Length of the range $e=1\%$ of the data universe side length



Both algorithms perform a single expansion of the network.

- RER first retrieves the candidate objects within the Euclidean range e and then expands the network
- RNE first expands and then performs the query on the data R-tree for the actual results.

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Summary

- Network distance is a more realistic metric than Euclidean distance.
- *Euclidean restriction* assumes the lower bounding property, which may not always hold in practice (if, for instance, the edge cost is defined as the expected travel time). On the contrary, *network expansion* permits a wide variety of costs associated with the edges.
- *Network expansion* has superior performance for nearest neighbour and range queries.



References

- Dimitris Papadias, Jun Zhang, Nikos Mamoulis, Yufei Tao: Query Processing in Spatial Network Databases. VLDB 2003:802-813
- A presentation by Afsin Akdogan in csci587 Fall'2010